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The Role of Regulation and Bank Competition in Small Firm Financing: Evidence from the Community Reinvestment Act

This paper analyzes how regulation that promotes greater access to bank credit, such as the Community Reinvestment Act (CRA), impacts the financing of small firms. It finds that when areas become CRA-eligible, the likelihood of bank lending to local small firms increases and firms reduce late trade credit payments, consistent with loans allowing small firms to pay trade credit more promptly and avoid late payment fees. The effect is more profound in low- and moderate-income areas where financial constraints are tighter due to low bank competition. The effect is also larger for small firms that operate in trade credit-dependent industries.

JEL codes: G14, G21, L13, L50

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DESPITE TECHNOLOGICAL INNOVATIONS IN LENDING, one-third of U.S. small firms report challenges in securing credit.¹ Consolidation of the banking industry may contribute to these firms' difficulties. Bank loans are small firms' most common source of external financing, but since the 2008 financial crisis, the average number of banks serving local areas has declined, especially in low-income areas. Mills, Battisto, and Lieberman (2019) report that while the number of banks per small firm is approximately the same in high-income and low-income areas, low-income areas have less than half as many banks and small firms.² Moreover, small firms in low-income areas are more likely to use relatively expensive nonbank sources of financing.

A decline in the number of banks is reason for concern because less bank competition reduces the formation of new firms (Black and Strahan 2002) and closures of local bank branches lead to a persistent decline in local small business lending (Nguyen 2019). Since an area's small firms are important providers of goods and services and are also employers of local residents, the economic prospects of low-income areas may be undermined.³ To expand credit availability and reduce the number of unbanked small firms and residents of low- and moderate-income (LMI) areas, policymakers have introduced programs to encourage bank lending to these communities.⁴ However, the impact on small firms targeted by these initiatives remains largely unexplored.

This study sheds light on the consequences of policy interventions on small firms by examining the effects of the Community Reinvestment Act (CRA), a U.S. regulation aimed at encouraging credit supply to LMI areas. First, we show that the areas that become newly eligible for CRA status experience a greater likelihood of a large increase in bank loans to small firms. Second, our analysis reveals that, on average, small firms in these newly CRA-eligible areas are less likely to delay their payments to vendors and suppliers, consistent with them using the liquidity from new bank loans to reduce overdue trade credit. Third, we find that the CRA's effect on small firms' improved trade credit performance is more pronounced in areas with high bank

1. See the Federal Reserve's 2020 Small Business Credit Survey, which was taken in the second half of 2019 prior to the onset of the COVID-19 pandemic.

2. Comparing U.S. zip code areas in the highest quartile of median family income (MFI) to areas in the lowest quartile of MFI, the ratio of banks to small firms is approximately 0.01 in both. However, in 2018, there was an average of 4.3 (2.0) banks in high- (low-) income areas, down from an average of 4.5 (2.3) banks in 2007. In contrast, in 2016, high- (low-) income areas had an average of 0.7 (1.2) providers of relatively expensive alternative financial services.

3. The U.S. Small Business Administration (SBA)'s 2019 *Small Business Profile* <https://cdn.advocacy.sba.gov/wp-content/uploads/2019/04/23142719/2019-Small-Business-Profiles-US.pdf>. Report estimates that there are 30.7 million small firms in the U.S. accounting for 99.9% of total businesses and 47.3% of total U.S. employment. Importantly, 37% of high-tech workers work for small firms.

4. The SBA, created in 1953, provides financing to young and growing small firms through several channels such as the 7(a) Loan Program and SBA 504 Certified Development Companies. The CRA, enacted in 1977, encourages depository institutions to meet the credit needs of their local LMI communities. More recently, the Federal Reserve established the Main Street Lending Program to support lending to small- and medium-sized organizations following the onset of the COVID-19 pandemic. This program terminated in January 2021.

concentration. In the absence of the CRA, such areas are likely to suffer from weak bank competition and more liquidity-constrained small firms. For small firms in industries that have traditionally relied on trade credit, there is a 3.4–5.4-day average reduction in the lateness of their payments relative to their creditors' invoice terms (due dates). There is also a 5.9–7.6% increase in the proportion of firms that are likely to avoid delinquency. In summary, our findings are consistent with CRA-eligibility having the most impact on small firms in areas with low bank competition, allowing them to significantly reduce trade credit delinquencies. Since penalties for late trade credit payments can be substantial (Petersen and Rajan 1994), the implied reduction in these firms' cost of financing is likely to promote economic development of their local area.

Our empirical tests use a unique data set constructed from a large-scale geocoding of National Establishment Time Series (NETS) and FDIC Summary of Deposits (SOD) information. The data set combines CRA loan data, small firm information, and bank branch information aggregated at the census tract level, which is the relevant geographic area for analyzing the effects of the CRA. The CRA divides census tracts into two groups: upper-middle tracts with MFI above 80% of the surrounding metropolitan area's MFI and low-moderate tracts with MFI below 80%. Bank regulators' examinations that determine CRA ratings place heavy weight on whether a bank with branches serving low-moderate tracts is meeting the credit needs of small firms and residents in those tracts.

Furthermore, we use the Dun & Bradstreet (D&B) PAYDEX score, a dollar-weighted indicator of trade payments, to measure the firm's promptness in paying its vendors and suppliers relative to the terms (due dates) of their invoices. The higher is a firm's PAYDEX score, the more prompt are its payments to trade creditors. Moreover, a lower PAYDEX score implies that a firm presents a higher risk of delinquency or default for its trade creditors (Kallberg and Udell 2003).

We use this data set to examine whether CRA-induced lending affects small firms' trade credit payments in conjunction with the market's level of bank competition, an issue that is largely absent from existing empirical studies. Our tests exploit the fact that CRA eligibility is based on a census tract's MFI ratio, which allows us to use a "fuzzy" regression-discontinuity (RD) design. The logic of the RD design is that the probability that a local bank's small firm lending responds to CRA incentives jumps discontinuously at the 80% MFI cutoff, inducing heterogeneity in a regulatory treatment that is unrelated to potential confounding characteristics. However, examiners rate a bank's CRA compliance based on a geographic assessment area that is broader than a single census tract. Thus, a tract's CRA eligibility may increase the probability that a bank lends to firms in that tract, but since it could still be in compliance by lending to other tracts, the probability need not be one. This probabilistic nature of a bank's tract-level compliance makes the fuzzy RD design more suitable compared to the sharp RD design that assumes deterministic tract-level compliance.

Our research contributes to a literature on bank competition and small firm financing. Berger et al. (1998) report that a decline in competition due to bank mergers leads to an initial shortfall in credit to small firms that is gradually eliminated by the entry

of new banks and increased lending by existing banks. Cetorelli and Strahan (2006) provide evidence that greater bank competition is associated with higher numbers of small firms, consistent with easier credit access lowering barriers to firm entry. Also, by promoting competition, deregulation of bank branching has raised the rate of new business incorporations (Black and Strahan 2002) and has increased small firms' access to bank loans at lower interest rates (Rice and Strahan 2010). Our findings show that government regulatory incentives to expand bank lending can also be effective in alleviating small firms' financing constraints, especially when bank competition is low.

Also relevant to our paper are studies, showing that markets for lending to small firms are local in the sense that the distance between a small firm and its potential bank lenders matters (Petersen and Rajan 2002, Brevoort, Holmes, and Wolken 2010). Compared to large corporations, there is often a lack of public "hard" information, such as audited financial statements, on small firms. Thus, banks often rely on the collection of private "soft" information through their relationships with small firms (Petersen and Rajan 1995), and obtaining such information is less costly when banks and firms are physically close. Thus, the distance between firms and their potential lenders often proxies as a measure of soft information collection or the strength of a banking relationship (Agarwal and Hauswald 2010). Our study suggests that when the distances between banks and small firms are larger because there are fewer banks in the local market, regulation can be effective in expanding access to bank credit.

Our paper also relates to research on trade credit. Firms facing impediments to bank credit are more likely to resort to trade credit (Petersen and Rajan 1994, 1997), which explains why the use of trade credit is greater in countries with weak banking institutions (Fisman and Love 2003). Lending to small firms is especially problematic due to their relative information opacity that requires more credit screening and monitoring from banks. Trade credit can alleviate this problem by incorporating in the lending relation the private information that suppliers have on their customers (Biais and Gollier 1997). However, while trade credit can be optimal for small firms from the viewpoint of transactions costs and liquidity, Petersen and Rajan shows that overdue trade credit may trigger additional charges that increase the cost of trade credit (also, Berger and Udell 1998).

In addition to pure financing, recent studies suggest that trade credit can be a mechanism for price discrimination whereby larger buyers with greater bargaining power obtain more favorable trade credit terms from smaller suppliers (Giannetti, Burkart, and Ellingsen 2011, Klapper, Laeven, and Rajan 2012). Thus, trade credit can be an instrument for a supplier to transfer surplus to a high-bargaining-power firm without affecting sales to other, lower-bargaining-power firms (Giannetti, Serrano-Velarde, and Tarantino 2021), thereby enabling firms to trade efficiently (Breza and Liberman 2017). On the other hand, extended trade credit terms imposed by larger buyers may force smaller, financially constrained suppliers to reduce investment and general expenditures (Murfin and Njoroge 2015) and may increase the likelihood that weaker suppliers face financial distress (Barrot 2016).

Our paper differs from the aforementioned studies by focusing on financially constrained small firms that, as buyers, are less likely to have substantial bargaining power over their suppliers. As such, our study extends the trade credit literature by showing that when banks ease financial constraints, small firms' trade credit performance improves. Our findings are also consistent with the empirical evidence that firms manage liquidity shortfalls by increasing the amount of trade credit drawn from suppliers (Amberg et al. 2021). We show that small firms also manage liquidity constraints through trade credit and that they are less likely to defer trade credit payments when these financial constraints abate due to bank lending. Finally, since deferred payments to suppliers increase the risk of losses to trade creditors that can propagate business failures in the economy (Jacobson and Von Schedvin 2015), our findings suggest that policies that encourage bank lending will ameliorate credit constraints that can have positive externalities by making trade credit chains more resilient.

Our paper also contributes to the more specialized literature examining the CRA's impact. Most research studies the CRA's effects on residential mortgages (Bostic and Robinson 2003, Dahl, Evanoff, and Spivey 2010, Bhutta 2011, Avery and Brevoort 2015, Ding and Nakamura 2020, Begley and Purnanandam 2021) finding mixed results regarding the effectiveness of the CRA on expanding mortgage lending. Fewer studies analyze the CRA's effects on lending to small firms. Zinman (2002) finds evidence that banks facing greater CRA incentives due to a 1995 CRA reform increased their lending to small firms but not in CRA-eligible LMI areas.⁵ In contrast, Bostic and Lee (2017) find that CRA-eligible census tracts experience more small firm lending than similar CRA-ineligible tracts during the periods 1996–2002 and 2012–2014, but not during 2003–2011. Ding, Lee, and Bostic (2018) analyze a 2013 adjustment in metropolitan area boundaries that led to an exogenous change in some census tracts' CRA eligibility. They find that a tract losing CRA eligibility led to a decline in lending to small firms, but there was not a statistically significant increase in lending to small firms when a tract gained eligibility. Kim, Lee, and Earle (2021) use 2004–2016 panel data and find that the number and amount of loans to small firms increase in tracts that become eligible for the CRA.

Research analyzing the CRA's effects on small firms has been confined to determining whether it changes the quantity of banks' small business loans. Our research goes a step further by addressing whether CRA-induced lending has benefited targeted small firms. This is an important issue for evaluating the economic efficacy of a government initiative. Our results document the conditions under which the CRA leads to an improvement in small firms' trade credit performance.

The paper is organized as follows. Our empirical methodology and data are discussed in Sections 1 and 2, respectively, while Section 3 presents the empirical findings. Section 4 concludes.

5. He also examined the counties where these banks operated and found evidence of a decline in personal bankruptcies but no change in employment, payrolls, or business bankruptcies.

1. EMPIRICAL METHODOLOGY

This section summarizes the CRA and discusses how an area's CRA eligibility is determined based on its LMI status and how our empirical tests exploit this eligibility.

1.1 *CRA Regulation*

The CRA, enacted in 1977, was motivated by the perception that many banks collected deposits from their branches located in LMI areas but extended insufficient credit to residents and small firms in these areas. Thus, the goal of the CRA is to expand bank lending to LMI areas beyond what banks would choose in the absence of this legislation. Banks are regularly examined and rated for CRA compliance by their federal regulators. These ratings are released to the public and carry consequences. First, federal regulators must consider a bank's CRA performance when evaluating its application for a merger or acquisition, for its formation of a branch, or for other business activity. Second, and more indirectly, community activists and public interest groups monitor banks and can impose public relations costs on those with poor CRA performance. Hence, they provide an independent source of bank discipline.

Given these penalties from noncompliance, one would expect that a bank's amount lent in a CRA-eligible market exceeds the amount lent in the same market if it was CRA-ineligible. Indeed, a Federal Reserve survey of banks finds that a majority engaged in some lending in CRA-eligible areas that they would not have done in the absence of the law (Avery, Bostic, and Canner 2005).

1.2 *Treatment of Census Tracts*

A bank's CRA rating is heavily weighted by its lending in census tracts that are designated as LMI areas. A census tract is an LMI area or "CRA-eligible" if its MFI is less than 80% of its surrounding metropolitan area's MFI based on information provided by the U.S. Census. According to the Federal Financial Institutions Examination Council (FFIEC), until the end of 2011, a tract's CRA eligibility was based on MFIs determined from year 2000 census data. From January 2012 forward, eligibility was based on MFIs using year 2010 census data. As a result, approximately 15% of the tracts that were marginally ineligible in 2011 using the 2000 census became CRA-eligible tracts in 2012 using the 2010 census.

Our tests compare these newly eligible census tracts relative to census tracts that remained ineligible using an RD design. Prior empirical studies analyzing the effects of the CRA on lending have also used an RD approach (see, e.g., Avery, Calem, and Canner 2003, Bhutta 2011, Avery and Brevoort 2015, Bostic and Lee 2017, Kim, Lee, and Earle 2021). However, unlike this prior work that uses a sharp RD design, we employ a fuzzy RD design. The difference is that we interpret the CRA's incentive on lending in an eligible census tract as probabilistic rather than deterministic. In other words, using the language of the RD literature, some banks are "noncompliant" in that they do not necessarily have an incentive to increase their lending to a particular

newly CRA-eligible census tract. Failing to consider the probabilistic nature of a loan increase induced by CRA eligibility means that the actual effect on the outcome variable is underestimated because all eligible tracts are equally counted, independent of whether the bank actually increased its lending.

The reason for a bank's possible noncompliance is that its overall CRA rating is based on an examination of its lending in a broader geographic assessment area that the bank can reasonably serve with its main office and branches. The implication is that a census tract's eligibility may increase the probability that a bank has a greater incentive to lend in that tract, but the bank could still obtain a satisfactory CRA rating by lending to other CRA-eligible tracts within a county or Metropolitan Statistical Area (MSA) where the bank has a presence. Consistent with our assumption that banks may be noncompliant in lending to small firms in CRA-eligible tracts is the Federal Reserve survey evidence reported in Avery, Bostic, and Canner (2005). Surveyed banks responded that they were significantly less likely to establish a special lending program in CRA-eligible areas for the case of small business loans compared to the case of residential mortgages.⁶

Therefore, we need to identify the eligible tracts that actually experience a CRA-induced increase in bank loan supply. This supply-driven loan growth is more easily observed among newly-eligible tracts compared to census tracts that have remained eligible based on the 2000 census data and thus had a more predictable supply of loans ever since. Focusing on newly-eligible tracts where banks are, presumably, less likely to have built long-term relationships with local firms helps to avoid any confounding effect from relationship lending on the firm's use of trade credit.

Our RD tests take the following form. Let X_i be census tract i 's ratio of MFI to the MFI of the surrounding metropolitan area. According to the CRA, loans to small firms in this tract become CRA eligible when $X_i < c = 80\%$, generating a discontinuity at the cutoff, c . We define as "treatment" a CRA-induced substantial growth in small firm loans, and we denote it with the dummy variable T_i . In sharp RD design, assignment to treatment coincides with the actual treatment, and thus, the jump in the probability of observing a substantial increase in small firm loans at the cutoff is from 0 to 1. As discussed above, CRA compliance is imperfect, calling for a fuzzy RD design where the probability of a substantial increase in small firm loans at the cutoff is less than 1. Yet, newly-eligible tracts are more likely to experience a substantial increase in small firm loan supply compared to census tracts that remain ineligible:

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} E[T_i | X_i = c - \epsilon] - \lim_{\epsilon \rightarrow 0} E[T_i | X_i = c + \epsilon] &= \lim_{\epsilon \rightarrow 0} \Pr(T_i = 1 | X_i = c - \epsilon) \\ &- \lim_{\epsilon \rightarrow 0} \Pr(T_i = 1 | X_i = c + \epsilon) > 0. \end{aligned} \quad (1)$$

6. Note that 54.9% of the respondents had set up a CRA Special Lending Program to expand residential mortgage lending, but only 10.4% of the respondents had set up such a program to expand small business lending. While banks could still expand lending without these programs, their existence indicates the strength of incentives to increase lending.

Now let Y_i denote an outcome variable for census tract i , such as a measure of how promptly small firms in census tract i pay their trade credit. There should be no reason, other than the eligibility of the tract, for this outcome variable to be a discontinuous function of the tract's MFI ratio. This discontinuity of Y_i is:

$$\lim_{\epsilon \rightarrow 0} E[Y_i | X_i = c - \epsilon] - \lim_{\epsilon \rightarrow 0} E[Y_i | X_i = c + \epsilon]. \quad (2)$$

Based on equations (1) and (2), the fuzzy RD effect of the eligibility-induced substantial increase in loans to small firms on the outcome variable Y_i is defined as:

$$\tau = \frac{\lim_{\epsilon \rightarrow 0} E[Y_i | X_i = c - \epsilon] - \lim_{\epsilon \rightarrow 0} E[Y_i | X_i = c + \epsilon]}{\lim_{\epsilon \rightarrow 0} E[T_i | X_i = c - \epsilon] - \lim_{\epsilon \rightarrow 0} E[T_i | X_i = c + \epsilon]} = \frac{\mu_{Y^-} - \mu_{Y^+}}{\mu_{T^-} - \mu_{T^+}}. \quad (3)$$

where $\mu_{U^\pm}$ is the limit of the expectation of variable U as X_i approaches c from below ($-$) or above ($+$).

1.3 Estimation Methodology

Estimation of the eligibility effect τ is performed using local polynomial estimators. Functions for T_i and Y_i above and below the cutoff are approximated by means of weighted polynomial regressions with weights computed by applying a kernel function on the distance of each observation to the cutoff point. We employ the robust local linear polynomial estimator of Calonico, Cattaneo, and Titiunik (2014), which corrects for the bias of bandwidth. Formally, the conventional local linear estimator for a positive bandwidth h , kernel function $K_h = K((X_i - c)/h)$, and a set of covariates $\mathbb{Z}_i$ is found by optimizing the equations:

$$\hat{\mu}_{U^-} = \arg \min_{\{b_0, b_1\}} \sum_n (\mathbb{1}(X_i < c)(U_i - b_0 - b_1(X_i - c) - \gamma \mathbb{Z}_i)^2 K_h), \quad (4)$$

$$\hat{\mu}_{U^+} = \arg \min_{\{b_0, b_1\}} \sum_n (\mathbb{1}(X_i \geq c)(U_i - b_0 - b_1(X_i - c) - \gamma \mathbb{Z}_i)^2 K_h), \quad (5)$$

where the random variable U_i is equal to either T_i or Y_i . Equations (4) and (5) use the recommended choice of local linear polynomial approximation, $b_0 + b_1(X_i - c)$, which has smaller asymptotic bias (Cattaneo, Idrobo, and Titiunik 2019). Moreover, in the analysis that follows, we employ the Epanechnikov kernel function, often called the optimal kernel, because it yields the lowest possible asymptotic mean squared error (MSE). In additional robustness tests, we also use the uniform kernel function.

The probability of treatment (e.g., a substantial small firm loan increase) is estimated by the local linear RD estimator that employs a local-quadratic bias correction:

$$\hat{\tau}_T = \hat{\mu}_{T^-} - \hat{\mu}_{T^+} - h_T^2 \mathbb{B}_T(h_T, b_T), \quad (6)$$

where we denote $\mathbb{B}_U(h_U, b_U)$ as the bias correction factor for variable U and b_U is the bias-correction bandwidth.⁷

Finally, the bias-corrected (and robust) local linear fuzzy RD estimator of the effect on the outcome variable (e.g., promptness paying trade credit) that uses a local quadratic estimate of the leading bias is:⁸

$$\hat{\tau}_Y = \frac{\hat{\mu}_{Y^-} - \hat{\mu}_{Y^+}}{\hat{\mu}_{T^-} - \hat{\mu}_{T^+}} - h_Y^2 \mathbb{B}_Y(h_Y, b_Y). \quad (7)$$

The choice of bandwidth, h , is consequential because, given the choices of local linear approximation and kernel function, the accuracy of the approximation is essentially determined by the bandwidth. For all point estimators of the RD effects, we use the recommended MSE optimal plug-in bandwidth h_{MSE} , which minimizes the asymptotic variance and bias of the estimator (Cattaneo, Idrobo, and Titiunik 2019). Additional robustness tests use the common MSE-optimal bandwidth selector for the sum of regression estimates bandwidth $h_{MSE,SUM}$, which is useful for minimizing overfitting. Furthermore, we use the CER optimal bandwidth h_{CER} , which minimizes the asymptotic coverage error rate of the bias-corrected confidence interval, for inference, in the falsification tests. Finally, we use the cluster at census tract level robust plug-in residuals variance estimator to compute the variance–covariance matrix.

2. DATA

This section discusses the data used in our tests and presents summary statistics of the sample. Our data come from three main sources.

2.1 *Small Business Loans, Census, and Demographic Data*

The FFIEC provides data on CRA lending and related census and demographic data. From it, we obtain information on small business loans made by individual banks. CRA regulations require that banks above given asset size thresholds report their commercial loans extended to small firms.⁹ We focus on loans originated to small firms with gross annual revenues of \$1 million or less. The reported annual loan data refer to new loans and include the total number of loans and the total loan amount at origination aggregated at the U.S. Census tract level. CRA small business

7. See Calonico, Cattaneo, and Titiunik (2014) for more details.

8. The bias-corrected and the robust point estimates are the same but the robust estimate's reported standard error is the cluster, plug-in residuals variance estimator.

9. Banks include national banks, state Federal Reserve member banks, state nonmember banks, and savings associations. The FFIEC reports the asset-size filing threshold each year, and it is typically over \$1 billion.

lending data cover roughly 86% of all loans of \$1 million or less (Greenstone, Mas, and Nguyen 2020).

The FFIEC also publishes census data at the tract level.¹⁰ Variables that pertain to our analysis include MFIs for each census tract and its surrounding MSA or Metropolitan Division (MD). As mentioned earlier, if a census tract is within a bank's assessment area, loans made in that tract qualify for CRA status based on the tract's MFI relative to the MFI in its MSA/MD.¹¹ Therefore, updates of tract MFI or MSA/MD MFI can change a tract's CRA eligibility. According to the FFIEC, 2011 MFI data for both census tracts and MSAs/MDs are based on the 2000 census, while 2012 MFI data are based on the 2010 census. Consistent with this income data, census tracts are classified based on the 2000 census for loans originated in 2011, while tracts are classified based on the 2010 census for loans originated in 2012. Additional demographic variables used in our analysis include the census tract's total population and its minority population as a percentage of total population.

2.2 *Small Firm Data*

Data on small firms are obtained from the National Establishment Time Series (NETS) compiled by Walls & Associates. Because small firms are mainly single establishments, we use the terms "firm" and "establishment" interchangeably. Among all NETS establishments, we exclude those from North American Industry Classification System (NAICS) industries of Agriculture, Forestry, Fishing, and Hunting whose CRA loans are reported separately, and from Education Services and Public Administration, which typically do not depend on small business bank loans. We further drop establishments with a missing NAICS code, data year, or employment numbers, and also inactive establishments such as those with a bankruptcy filing indicator. Following the literature of the NETS database (Barnatchez, Crane, and Decker 2017), we also exclude establishments reporting only one employee, that is, self-employed. We consider an establishment to be a small firm based on the definition used by the Small Business Administration (SBA), namely, a manufacturing industry firm with less than 500 employees and a nonmanufacturing industry firm with 3-year average annual sales below \$7.5 million.

NETS also reports each firm's PAYDEX score, which is compiled by Dun & Bradstreet (D&B) based on reports from a firm's vendors and suppliers. D&B uses a rolling, 12-month record of a firm's trade payments to compute a value-weighted indicator of the timeliness of its payments relative to its bills' terms (due dates). PAYDEX ranges from 1 to 100, where a higher number indicates that a firm is more prompt in paying its trade credit. Specifically, a score of 100 indicates a payment

10. A census tract, which is a relatively permanent subdivision of a county, generally contains between 1,200 and 8,000 inhabitants and varies in size across urban and rural areas.

11. For tracts located outside a MSA/MD, the surrounding area MFI is the statewide non-MSA/MD MFI.

anticipates (prepays) an invoice, while 90 indicates a payment during an invoice's early-payment discount period.¹² A score of 80 indicates that a payment is received on terms (by its invoice due date), while scores of 70, 60, 50, 40, 30, 20, and 0–19 indicate that a payment is received beyond terms by 15, 22, 30, 60, 90, 120, and over 120 days, respectively. The PAYDEX score is then computed as a dollar-weighted average of a firm's individual payment scores.¹³ As might be expected, Kallberg and Udell (2003) find that a low PAYDEX score is a strong predictor of firm failure.

To assign establishments to census tracts, we perform geocoding from an establishment's latitude and longitude provided by NETS to determine within which tract's boundary it falls, where tract boundary files are obtained from the U.S. Census website. Consistent with CRA lending data, NETS data before 2012 are mapped to census tracts based on the 2000 census map, while 2012 NETS data are mapped based on 2010 census map.¹⁴ For each year, we calculate the total number of small firms and their average PAYDEX scores per census tract. In addition, we calculate the proportion of firms with PAYDEX scores higher than 70. We choose to focus on the threshold of 70 for two reasons. First, this threshold is approximately equal to the average PAYDEX score observed among small firms in our sample.¹⁵ Second, a firm with a score below 70 is, on average, paying later than a bill's terms and is more likely to be missing out on early-payment discounts and subject to late-payment penalties.¹⁶ As such, small firms with a score below 70 are likely resorting to relatively expensive trade credit.

2.3 Bank Branch Data

Bank branch data come from the FDIC Summary of Deposits (SOD), the annual report of branch office deposits for all FDIC-insured institutions including insured U.S. branches of foreign banks. Following standard treatment, we keep only brick-and-mortar branches and remove any branches with missing or duplicated physical

12. Many trade creditors provide "two-part" contracts that allow for discounts (Ng, Smith, and Smith 1999). Based on the 1998 National Survey of Small Business Finance (NSSBF), Cunnat and Garcia-Appendini (2012) report that the most common is "2/10, net 30," which means that the bill is due within 30 days but a discount of 2% is applied if paid within 10 days. Also from the 1998 NSSBF, Cunnat (2006) finds that approximately 51% of the main suppliers to small firms offer a discount.

13. See <https://docs.dnb.com/static/doc-uploads/supplier/en-US/support/FAQs.pdf> for more details on calculating PAYDEX.

14. We exclude establishments that moved during our sample period to avoid potential confounding factors that led a firm to relocate to obtain a CRA loan. We also drop establishments with missing coordinates.

15. Only 0.5% of firms have a PAYDEX score of 80 or above, 73.1% have a score between 70 and 80, and 26.4% have a score of 70 or below.

16. According to Cunnat (2006) and Cunnat and Garcia-Appendini (2012), the 1998 NSSBF shows that 46% of small firms forgo taking an early-payment discount and 57% of small firms' trade credit contracts have an explicit late-payment penalty.

addresses. The SOD reports latitudes and longitudes of most physical branches, and for those that are missing this data, we impute coordinates from geocoding the branch addresses.

To estimate the relevant bank branches to which a small firm in a given census tract would likely have access, we draw a circle with a search radius around the population center of each census tract.¹⁷ We assume a small firm searches for bank branches within 2 miles around the center of a tract in an urban area, 5 miles in a mixed urban/rural area, and 10 miles in a rural area.¹⁸ Any bank branch having coordinates within the relevant search radius from the tract center is considered to serve small firms in that tract. Thus, we are able to identify for each year all bank branches serving a particular census tract.

We calculate the Herfindahl–Hirschman Index (HHI) of banks serving each tract by estimating their market shares of deposits in branches serving that tract.¹⁹ We also calculate separately the total number of branches of large banks, defined as banks with assets exceeding the annual CRA regulatory size threshold, and the total number of branches of small banks, defined as banks with assets below that threshold. The large versus small bank distinction is potentially important since prior research suggests that small banks are more inclined to fund local small firms (Berger et al. 2005). In addition, small banks below the CRA regulatory threshold are subject to fewer reporting requirements.

2.4 Summary Statistics

Our sample is composed of observations at the census tract level during the years 2011 and 2012. It includes two types of census tracts: the newly CRA-eligible tracts after the incorporation of the 2010 census data and the tracts that remained CRA-ineligible after the census data update. Moreover, we exclude all new tracts in the 2010 census that were not included in the 2000 census and also restrict the analysis to tracts whose boundaries were relatively unchanged. Overall, we have 73,366 observations with 69,315 observations belonging to upper-middle-income (UMI) tracts with an MFI above the 80% cutoff and 4,051 observations belonging to LMI tracts with a MFI below the 80% cutoff.

17. Tract population center files are downloaded from the census website (<https://www.census.gov/geographies/reference-files/time-series/geo/centers-population.2010.html>).

18. Using the same distance across sparsely and densely populated areas assumes that it takes the same amount of time to cover a distance of 10 miles in rural and urban areas, which is counterfactual. Consequently, a fixed distance would inflate the number of branches in more densely populated areas while underestimating the figure for rural areas. The same approach was adopted by the Bank Policy Institute's report *Some Facts about Bank Branches and LMI Customers*.

19. Any branches of the same bank serving a tract are treated as one unit because they are not considered competitors.

Table 1, Panel A, provides summary statistics on our bank data for the subsample of UMI census tracts and the subsample of LMI census tracts. It reports statistics on the number and amount of small business loans, the number of branches split by bank size, and the HHI bank concentration measure. Table 1, Panel B, summarizes the demographic data that include the tract population, the population's percentage of minorities, and the MFI. Moreover, it includes business data, specifically the number of small firms, the average PAYDEX score, and the proportion of firms with a PAYDEX score higher than 70.

A direct comparison between the two groups of tracts reveals that LMI tracts, on average, are slightly less populated and have a significantly higher proportion of minority population compared to UMI tracts. Consistent with Mills, Battisto, and Lieberman (2019), on average, LMI tracts have fewer small firms, fewer large and small bank branches, and lower bank competition (higher HHI) compared to UMI tracts. Given these statistics, it is not surprising that, on average, LMI tracts have lower numbers and amounts of small business loans and lower firm PAYDEX scores compared to the UMI tracts.

Our baseline RD estimates use tract observations from Table 1 but assign greater weights to those observations close to the 80% MFI cutoff. To compare the subsample of tracts within 10% of this cutoff, Panels A and B of Table 2 summarize the characteristics of tracts with a MFI between 80% and 90% to tracts with an MFI between 70% and 80%. The table indicates that differences between UMI and LMI tracts close to the cutoff are mostly eliminated, allowing us to conclude that these tracts have similar demographic, business, and banking characteristics.

As a robustness exercise, we estimate the effects of CRA eligibility using a separate sample of matched tracts. At first glance, the use of a matched sample in RD design seems unnecessary. Heckman, LaLonde, and Smith (1999) argue that RD estimators constitute a special case of selection on observables, and therefore, the RD estimator is a limit form of matching. Indeed, as we observed above, the heterogeneity of observable tract characteristics is largely eliminated for tracts with MFI ratios close to the CRA cutoff. Nonetheless, tracts from different states may be subject to different regulatory and legal frameworks, such as branch banking policies. Since these disparities could affect the supply of bank credit, we also perform an alternative estimation exercise using a matched sample that accounts for unobservable state-induced heterogeneity.

Specifically, for every tract that becomes eligible for CRA regulation, we find a matched tract that remained ineligible and that is located in the same MSA or in the same state if the tract is not in an MSA. Matching is performed based on the tracts having similar population, proportion of minority population, MFI, number of small firms, number of large bank branches, and number of small bank branches. Matching uses a nearest neighbor approach with distances that are a weighted function of the above covariates. Weights are derived from the Mahalanobis method, which is based on the inverse of the covariates' variance-covariance matrix. Panels A and B of

TABLE 1
DESCRIPTIVE STATISTICS FOR UPPER-MIDDLE-INCOME TRACTS & LOWER-MODERATE-INCOME TRACTS

Panel A: Bank branch and small business loan data									
Variables	80% ≤ MFI (N = 69,315)				MFI > 80% (N = 4,051)				
	mean	sd	min	max	mean	sd	min	max	
Number of small business loans	35.25	33.72	0	1,146	24.17	31.72	0	1,191	
Total amount (\$Th) of small business loans	1,121	1,388	0	33,090	848.9	1,290	0	35,015	
Branches of small banks	2,310	2,751	0	48	1,987	2,339	0	29	
Branches of large banks	5,829	4,997	0	62	5,333	4,181	0	49	
HHI-tract	3,099	2,185	485.5	10,000	3,257	2,162	675.1	10,000	
Panel B: Demographic and small business data									
Variables	80% ≤ MFI (N = 69,315)				MFI > 80% (N = 4,051)				
	mean	sd	min	max	mean	sd	min	max	
Population	4,314	1,747	23	33,041	4,138	1,747	37	16,326	
Minority population (%)	22.36	22.17	0	99.93	42.68	28.57	1.030	99.87	
Tract MFI	68,841	28,827	20,833	250,001	45,564	10,789	2,499	85,912	
Number of small firms	182.8	162.4	1	5,774	158.4	171.2	5	5,515	
PAYDEX score	71.06	3,462	33.38	80.57	69.55	4,169	18	80	
Proportion of firms with PAYDEX higher than 70	0.753	0.0883	0	1	0.719	0.100	0	1	

TABLE 2
DESCRIPTIVE STATISTICS FOR UPPER-MIDDLE-INCOME TRACTS & LOWER-MODERATE-INCOME TRACTS WITH MFI CLOSE TO THE CRA THRESHOLD

Panel A: Bank branch and small business loan data										
Variables	80% ≤ MFI < 90% (N = 10,863)					70% > MFI > 80% (N = 2,608)				
	mean	sd	min	max		mean	sd	min	max	
Number of small business loans	24.68	23.39	0	479		24.93	34.45	0	1,191	
Total amount (\$Th) of small business loans	825.0	1,116	0	20,302		857.3	1,353	0	35,015	
Branches of small banks	2,032	2,347	0	48		1,974	2,212	0	20	
Branches of large banks	4,929	4,302	0	62		5,258	4,108	0	49	
HHI	3,434	2,322	620.3	10,000		3,238	2,164	675.1	10,000	

Panel B: Demographic and small business data

Variables	80% ≤ MFI < 90% (N = 10,863)					70% > MFI < 80% (N = 2,608)				
	mean	sd	min	max		mean	sd	min	max	
Population	4,153	1,612	37	16,532		4,247	1,722	494	16,326	
Minority population (%)	29.47	26.57	0	99.76		39.09	28.13	1.040	99.67	
Tract MFI	47.805	11,267	20,833	95,777		48,726	10,303	24,274	85,912	
Number of small firms	154.0	127.9	1	4,024		161.4	183.4	8	5,515	
PAYDEX score	70.37	3,808	33.38	80.19		69.76	4,036	18	79.34	
Proportion of firms with PAYDEX higher than 70	0.734	0.0965	0	1		0.726	0.0966	0	1	

Table 3 summarize the characteristics of the same-state matched tract-year observations split between UMI and LMI tracts.²⁰

3. EMPIRICAL RESULTS

3.1 CRA Regulation and Bank Lending to Small Firms

The fundamental assumption of this study's fuzzy RD design is that the probability of a substantial increase in bank lending to small firms is greater in census tracts that become CRA-eligible compared to census tracts that remain ineligible. We begin the empirical investigation by providing explicit evidence in support of this assumption. In particular, we examine if there is an increase of at least 20% either in the number or in the total amount of a census tract's loans to small firms.²¹

We begin by plotting the probability of such a loan increase relative to the ratio of the census tract's MFI ratio using the evenly spaced method to select the number of bins. We do so separately for low and high bank competition census tracts. We follow Department of Justice (DoJ) guidelines and define low bank competition tracts as having an HHI greater than 1800 and high bank competition tracts as having an HHI less than 1800.²² RD plots in Figure 1 reveal that there is a discontinuous jump at the MFI cutoff. In particular, for both low and high bank competition census tracts, the probability of a substantial loan increase to small firms is significantly higher for census tracts with an MFI ratio lower than 80% compared to census tracts with an MFI ratio higher than 80%.

Columns (1) and (2) in Table 4 present the estimated discontinuity jump in the probability of a substantial increase in loans to small firms, $\hat{\tau}_T = \mu_{T-} - \mu_{T+}$, for the unmatched and the matched samples, respectively. These estimations use a set of covariates $\mathbb{Z}_i$ that includes the census tract's population, the population's proportion of minorities, the MFI, and the numbers of small and large bank branches within the tract's search area. The estimates indicate that CRA eligibility leads to approximately 25% and 34% higher probabilities of a substantial increase in loans for the unmatched and matched samples, respectively. These results are consistent with prior evidence that the CRA promotes small business lending. Ding, Lee, and Bostic (2018) show that the CRA leads to an increase in the number of loan originations and Kim, Lee, and Earle (2021) find a 3–04% increase in the number of small business loans and a 6–10% increase in the dollar loan amount in tracts that become CRA-eligible.

20. The number of LMI and UMI tracts differs in Table 3 since, in 2011, all tracts were ineligible UMI tracts, while 3,974 of these tracts became eligible LMI tracts in 2012.

21. For the entire population of tracts, the median annual growth is 1.3% in the number of loans to small firms and 0.06% in the amount. The chance of observing at least 20% annual growth in the number of loans is 25%. We have tested other thresholds of loan growth in the range between 15% and 25% and all reported results remain statistically significant.

22. More details are available at <https://www.justice.gov/atr/bank-merger-competitive-review-introduction-and-overview-1995>.

TABLE 3
DESCRIPTIVE STATISTICS FOR MATCHED UPPER-MIDDLE-INCOME TRACTS AND LOWER-MODERATE-INCOME TRACTS

Panel A: Bank branch and small business loan data										
Variables	80% ≤ MFI (N = 10,274)					MFI > 80% (N = 3,974)				
	mean	sd	min	max		mean	sd	min	max	
Number of small business loans	25.59	29.66	0	1,146		24.07	31.85	0	1,191	
Total amount (\$Th) of small business loans	862.2	1,214	0	32,205		847.1	1,290	0	35,015	
Branches of small banks	2.147	2.441	0	32		1.986	2.346	0	29	
Branches of large banks	5.583	4.268	0	50		5.350	4.189	0	49	
HHI	3,092	2,048	649.9	10,000		3,253	2,164	675.1	10,000	

Panel B: Demographic and small business data

Variables	80% ≤ MFI (N = 10,274)					MFI > 80% (N = 3,974)				
	mean	sd	min	max		mean	sd	min	max	
Population	4,093	1,534	91	15,955		4,138	1,751	37	16,326	
Minority population (%)	33.26	26.84	0.440	99.77		42.94	28.58	1.030	99.87	
Tract MFI	51,150	13,629	20,833	200,001		45,322	10,668	2,499	85,912	
Number of small firms	158.3	166.6	5	5,774		157.9	171.8	5	5,515	
PAYDEX score	70.09	3.897	33.38	80.19		69.55	4.179	18	80	
Proportion of firms with PAYDEX higher than 70	0.725	0.0987	0	1		0.719	0.101	0	1	

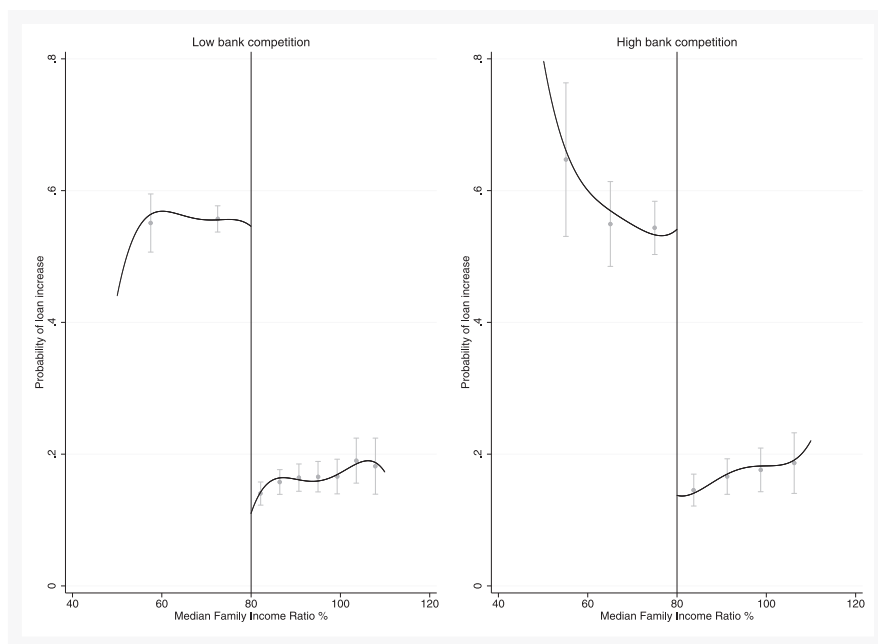


Fig. 1. Probabilities of a Substantial Increase in Loans to Small Firms.

Columns (3)–(6) of Table 4 also present estimates of the probability of a substantial loan increase due to CRA eligibility for subsamples based on a census tract's level of bank concentration. For the unmatched (matched) samples, CRA eligibility leads to an approximately 27% (37%) higher probability of a substantial increase in loans in low competition tracts and a 24% (33%) higher probability in high competition tracts. In summary, the evidence indicates a significant rise in the probability that lending to small firms increases in newly-eligible census tracts, and this rise occurs in tracts with both low and high bank competition. The latter result is important since it would be possible that banks comply with the CRA only in low competition areas where they enjoy higher rents (Petersen and Rajan 1995). In that were true, variation in the impact of the CRA on small firms by a tract's bank concentration could be attributed to banks' different compliance decisions. Based on Figure 1 and Table 4, columns (3)–(6), we can safely dismiss such a possibility.

3.2 CRA Regulation and Small Firm Trade Credit

The CRA-induced increase in bank lending documented in the previous section does not automatically imply that small firms will alter their trade payment performance. We now address this issue by providing empirical evidence that when a census tract first attains CRA status, small firms' average PAYDEX score rises,

TABLE 4
THE CRA'S EFFECT ON SMALL FIRM LOAN GROWTH

	All tracts		Low bank competition tracts		High bank competition tracts	
	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Conventional	0.246 ^{***} (0.0197)	0.341 ^{***} (0.0212)	0.266 ^{***} (0.0215)	0.361 ^{***} (0.0235)	0.231 ^{***} (0.0435)	0.321 ^{***} (0.0452)
Bias-corrected	0.250 ^{***} (0.0197)	0.344 ^{***} (0.0212)	0.271 ^{***} (0.0215)	0.367 ^{***} (0.0235)	0.239 ^{***} (0.0435)	0.327 ^{***} (0.0452)
Robust	0.250 ^{***} (0.0230)	0.344 ^{***} (0.0247)	0.271 ^{***} (0.0249)	0.367 ^{***} (0.0270)	0.239 ^{***} (0.0513)	0.327 ^{***} (0.0528)
Observations	73,366	14,248	49,525	10,298	20,957	3,544
Robust 95% CI	[0.205; 0.295]	[0.296; 0.393]	[0.222; 0.32]	[0.314; 0.419]	[0.138; 0.339]	[0.223; 0.43]
Kernel Type	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
BW Type	mserd	mserd	mserd	mserd	mserd	mserd
Sample	Total	Matched	Total	Matched	Total	Matched
Order Loc.Poly.(p)	1,000	1,000	1,000	1,000	1,000	1,000
Order Bias(q)	2,000	2,000	2,000	2,000	2,000	2,000
BW Loc.Poly.(h)	11,306	10,124	12,672	11,149	11,386	10,680
BW Bias (b)	19,342	17,823	22,558	20,808	18,938	18,249

NOTE: The outcome variable is the probability of an increase of at least 20% in either the census tract's number or the total amount of bank loans made to small firms. Covariates include the tract's population, the population's proportion of minorities, the MFI, and the numbers of small and large bank branches. Bank competition is measured by HHI. The Conventional estimator of RD is based on the first-order (linear) polynomial estimators in (4) and (5). The bias-corrected and robust estimator is the first-order polynomial RD estimator in (6). Reported bandwidths are percentage points of the MFI ratio. Standard errors in parentheses. Significance levels are ***1%, **5%, and *10%.

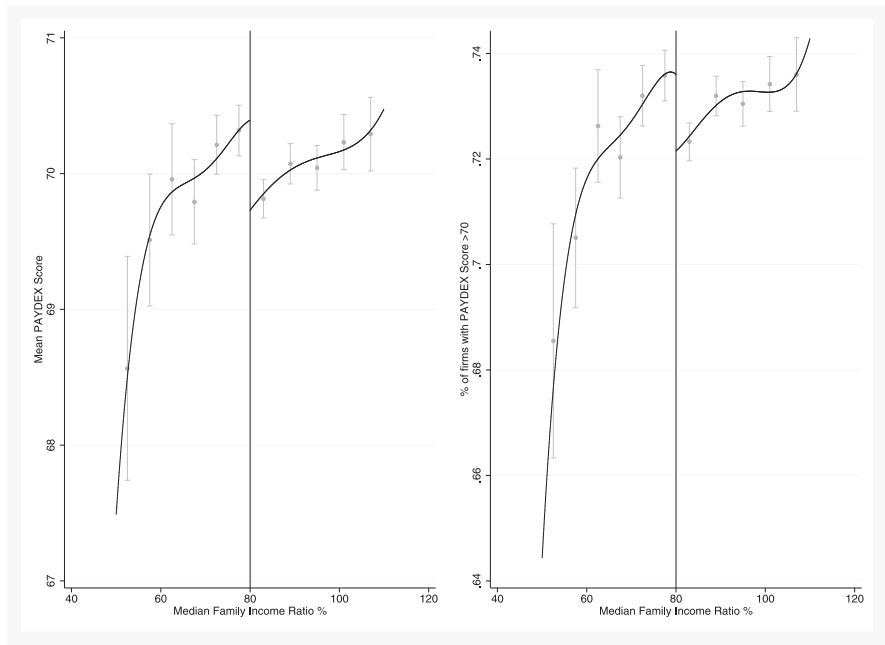


Fig. 2. Discontinuity in Trade Credit Scores at the 80% MFI Ratio.

consistent with them using bank credit to pay vendors and suppliers on time and avoiding delinquency-related charges that would make trade credit expensive.

As a preliminary analysis, we begin by plotting the PAYDEX score distribution 1 year after potential CRA eligibility relative to the ratio of the census tract's MFI ratio using the evenly spaced method to select the number of bins. The left RD plot in Figure 2 reveals that there is a discontinuous jump at the 80% MFI cutoff. In particular, small firms located in census tracts with an MFI ratio lower than 80% have an average PAYDEX score, 1 year after the tract's eligibility, significantly higher than the average score of small firms located in census tracts with an MFI ratio higher than 80%. Moreover, the lower value of the 95% confidence interval for the sample mean of the PAYDEX score within the bin close to, but below, the cutoff is higher than the upper value of the 95% confidence interval for the sample mean of the PAYDEX score within the bin close to, but above, the cutoff, suggesting that the jump is statistically significant at the 5% level. In addition to the discontinuity at the cutoff, note that the plot has an upward trend, indicating that small firms located in census tracts with higher MFI ratios tend to have higher PAYDEX scores. Similar conclusions are drawn by looking at the rightmost plot in Figure 2, which gives the RD plot of the proportion of firms with a PAYDEX score of more than 70.

We now turn to estimating the effect of CRA eligibility, $\hat{\tau}_Y$ in equation (7), on small firms' average PAYDEX score 1 year after their census tract's eligibility. The

TABLE 5
THE CRA'S EFFECT ON SMALL FIRMS' TRADE CREDIT PERFORMANCE

Panel A: PAYDEX score				
Variables	(1)	(2)	(3)	(4)
Conventional	1.413** (0.583)	1.434** (0.568)	1.683*** (0.484)	1.772*** (0.495)
Bias-corrected	1.308** (0.583)	1.400** (0.568)	1.698*** (0.484)	1.719*** (0.495)
Robust	1.308* (0.676)	1.400** (0.635)	1.698*** (0.559)	1.719*** (0.560)
Observations	73,366	73,366	14,248	14,248
Robust 95% CI	[−0.018; 2.633]	[0.155; 2.645]	[0.602; 2.794]	[0.622; 2.816]
Kernel Type	Epanechnikov	Uniform	Epanechnikov	Uniform
BW Type	mserd	msecomb1	mserd	msecomb1
Sample	Total	Total	Matched	Matched
Order Loc.Poly.(p)	1.000	1.000	1.000	1.000
Order Bias(q)	2.000	2.000	2.000	2.000
BW Loc.Poly.(h)	10.950	10.256	10.586	9.257
BW Bias (b)	19.221	21.994	19.065	19.138

Panel B: Proportion of firms with PAYDEX > 70				
Variables	(1)	(2)	(3)	(4)
Conventional	0.0303** (0.0152)	0.0347** (0.0145)	0.0357*** (0.0127)	0.0389*** (0.0129)
Bias-corrected	0.0275* (0.0152)	0.0331** (0.0145)	0.0353*** (0.0127)	0.0387*** (0.0129)
Robust	0.0275 (0.0176)	0.0331** (0.0165)	0.0353** (0.0147)	0.0387*** (0.0144)
Observations	73,366	73,366	14,248	14,248
Robust 95% CI	[−0.007; 0.062]	[0.001; 0.066]	[0.006; 0.064]	[0.01; 0.067]
Kernel Type	Epanechnikov	Uniform	Epanechnikov	Uniform
BW Type	mserd	msecomb1	mserd	msecomb1
Sample	Total	Total	Matched	Matched
Order Loc.Poly.(p)	1.000	1.000	1.000	1.000
Order Bias(q)	2.000	2.000	2.000	2.000
BW Loc.Poly.(h)	11.053	10.576	9.990	8.828
BW Bias (b)	19.586	20.734	18.010	19.070

NOTE: The outcome variables are the 1-year forward tract-level average PAYDEX score of small firms (Panel A) and the proportion of firms with a PAYDEX score exceeding 70 (Panel B). Covariates include the tract's population, the population's proportion of minorities, the MFI, and the numbers of small and large bank branches. The conventional estimator of RD is based on the first-order (linear) polynomial estimators in (4) and (5). The bias-corrected and robust is the first-order polynomial fuzzy RD estimator in (7). Reported bandwidths are percentage points of the MFI ratio. Standard errors in parentheses. Significance levels are ***1%, **5%, and *10%.

estimation again includes covariates for the census tract's population, the population's proportion of minorities, the MFI, and the numbers of small and large bank branches within the search area. The results in Table 5, Panel A, show that there is a statistically significant increase in the average PAYDEX score of small firms in the year following CRA eligibility. For example, the bias-corrected local linear estimate varies from 1.31 to 1.40 points for the full sample and 1.70 to 1.72 points for the matched sample. The magnitude of the effect is equal to half of the score's standard deviation. Based on how PAYDEX is calculated, a one-point improvement at the sample average PAYDEX of 70 implies that, relative to invoice terms, a firm pays its trade credit earlier by

1.5 days.²³ Thus, the bias-corrected estimates imply that CRA eligibility reduces the lateness of small firms' trade credit payments relative to invoice terms by an average of 2.0–2.6 days.

Table 5, Panel B, provides evidence similar to that of Panel A by showing that in newly CRA-eligible census tracts, there is an increase in the proportion of small firms having a PAYDEX score exceeding 70. The effect measured by the bias-corrected local linear estimate of $\hat{\tau}_Y$ in equation (7) varies from 2.75% to 3.87%. In summary, this section's evidence is consistent with small firms in newly CRA-eligible census tracts using bank loans to improve their trade credit payment performance.

3.3 CRA Regulation, Bank Competition, and Small Firm Trade Credit

The effect of CRA regulation on small firms' trade credit documented above applies to our sample's average census tract. In this section, guided by the intuition that local markets underserved by banks are more likely to benefit from CRA eligibility, we examine this effect conditional on a tract's bank competition. We begin by plotting the PAYDEX score distribution 1 year after potential CRA eligibility versus the census tract's MFI ratio for low and high bank competition tracts. For low competition tracts having an HHI above 1800, the top left RD plot in Figure 3 indicates a discontinuity in the average PAYDEX score at the MFI cutoff. The 95% confidence interval's lower bound for the average PAYDEX score close to, but below, the cutoff is higher than the upper bound of the 95% confidence interval for the average PAYDEX score close to, but above, the cutoff, suggesting that the jump is statistically significant at the 5% level. Similarly, the bottom left RD plot in Figure 3 shows that in low competition tracts, there is also a statistically significant jump in the proportion of firms with a PAYDEX score exceeding 70 at the MFI cutoff.

The right RD plots in Figure 3 show the distribution of the average PAYDEX score (top panel) and the proportion of firms with PAYDEX exceeding 70 (bottom panel) for census tracts with high bank competition having an HHI below 1800. While these top and bottom right plots reveal a discontinuous jump at the 80% MFI cutoff for CRA eligibility, the discontinuity does not appear statistically significant because the two confidence intervals, just below and just above the cutoff, overlap.

Table 6 provides formal estimates of the effects of CRA eligibility on PAYDEX scores for unmatched and matched samples split between low versus high bank competition tracts. Its results reinforce the graphical findings in Figure 3. Columns (1) and (2) of Table 6, Panel A, show that in tracts with low bank competition ($HHI > 1800$), CRA eligibility has a strong positive effect on the average PAYDEX score 1 year after becoming eligible. As a robustness check, columns (3) and (4) of Table 6, Panel A, present results using an alternative measure of low bank competition, namely, census tracts where the total number of bank branches is below the overall sample median.

23. Since a PAYDEX of 80 represents a payment at an invoice's terms (due date) while a PAYDEX of 70 represents a payment at 15 days beyond terms, a one-point increase in PAYDEX over the range from 70 to 80 is equivalent to a 1.5 day reduction in the lateness of payments relative to terms.

TABLE 6
BANK COMPETITION AND THE CRA'S EFFECT ON SMALL FIRMS' TRADE CREDIT PERFORMANCE

Variables	Panel A: PAYDEX score							
	Low bank competition tracts				High bank competition tracts			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Conventional	1.524*** (0.589)	1.692*** (0.564)	1.839*** (0.704)	2.117*** (0.573)	2.071* (1.221)	2.128** (1.037)	1.468* (0.800)	1.539** (0.686)
Bias-corrected	1.468** (0.589)	1.677*** (0.564)	1.719** (0.704)	2.128*** (0.573)	2.007 (1.221)	2.093** (1.037)	1.426* (0.800)	1.435** (0.686)
Robust	1.468** (0.679)	1.677** (0.652)	1.719** (0.827)	2.128*** (0.661)	2.007 (1.429)	2.093* (1.205)	1.426 (0.927)	1.435* (0.788)
Observations	49,525	10,298	40,290	8,187	20,957	3,544	33,076	6061
Robust 95% CI	[0.137; 2.8]	[0.399; 2.955]	[0.097; 3.34]	[0.833; 3.422]	[-0.795; 4.808]	[-0.268; 4.455]	[-0.39; 3.243]	[-0.11; 2.979]
Kernel Type	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
BW Type	mserd	mserd	mserd	mserd	mserd	mserd	mserd	mserd
Sample	Total	Matched	Total	Matched	Total	Matched	Total	Matched
Order Loc.Poly.(p)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Order Bias(q)	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
BW Loc.Poly.(h)	12.898	9.605	12.288	13.463	12.765	11.221	16.133	12.895
BW Bias (b)	23.545	17.434	20.851	24.959	21.595	19.387	29.246	23.780

(Continued)

TABLE 6
(CONTINUED)

Variables	Panel B: Proportion of firms with PAYDEX > 70							
	Low bank competition tracts				High bank competition tracts			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Conventional	0.0358** (0.0152)	0.0349** (0.0140)	0.0340* (0.0185)	0.0369** (0.0160)	0.0368 (0.0312)	0.0362 (0.0264)	0.0369* (0.0209)	0.0361* (0.0185)
Bias-corrected	0.0341** (0.0152)	0.0348** (0.0140)	0.0290 (0.0185)	0.0341** (0.0160)	0.0351 (0.0312)	0.0351 (0.0264)	0.0367* (0.0209)	0.0377** (0.0185)
Robust	0.0341* (0.0175)	0.0348** (0.0160)	0.0290 (0.0217)	0.0341* (0.0187)	0.0351 (0.0367)	0.0351 (0.0306)	0.0367 (0.0243)	0.0377* (0.0212)
Observations	49,525	10,298	40,290	8,187	20,957	3,544	33,076	6,061
Robust 95% CI	[0 ; 0.068]	[0.003 ; 0.066]	[-0.013 ; 0.072]	[-0.003 ; 0.071]	[-0.037 ; 0.107]	[-0.025 ; 0.095]	[-0.011 ; 0.084]	[-0.004 ; 0.079]
Kernel Type	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
BW Type	mserd	mserd	mserd	mserd	mserd	mserd	mserd	mserd
Sample	Total	Matched	Total	Matched	Total	Matched	Total	Matched
Order Loc. Poly.(p)	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Order Bias(q)	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000
BW Loc. Poly.(h)	13,508	10,328	11,992	10,745	12,532	11,521	15,813	11,886
BW Bias (b)	24,633	19,145	20,308	18,457	21,011	19,855	28,387	21,584

NOTE: The outcome variables are the 1-year forward tract-level average trade credit score of small firms (Panel A) and the proportion of small firms with a trade credit score of more than 70 (Panel B). Covariates include the tract's population, the population's proportion of minorities, the MFI, and the numbers of small and large bank branches. Bank competition is measured by HH (above versus below 1800) in (1)-(2) and (4) and (6) by the number of bank branches (below versus above the median) in (3)-(4) and (7)-(8). The conventional estimator of RD is based on the first-order (linear) polynomial estimators in (4) and (5). The bias-corrected and robust is the first-order polynomial fuzzy RD estimator in (7). Reported bandwidths are percentage points of the MFI ratio. Standard errors in parentheses. Significance levels are ***1%, **5%, and *10%.

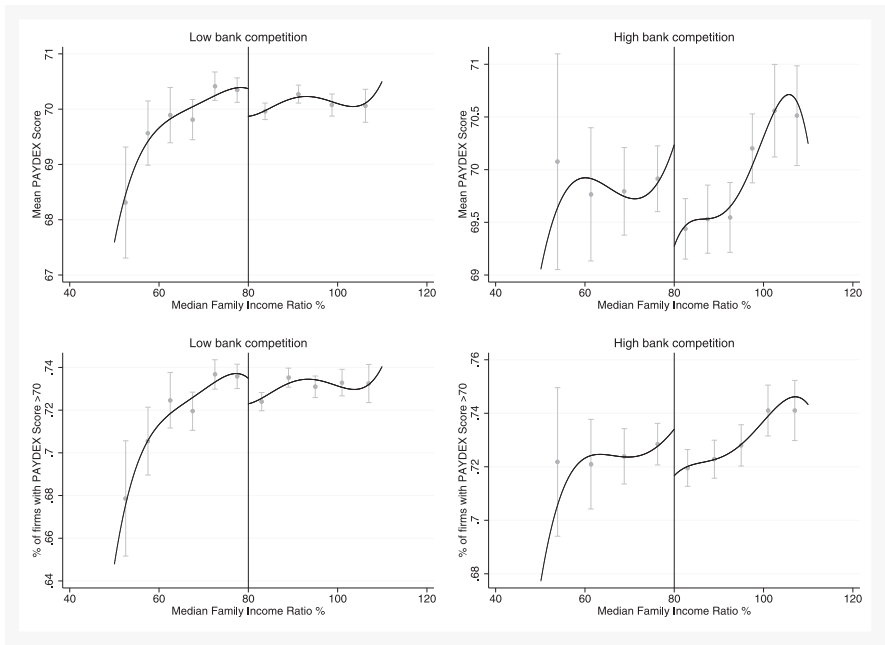


Fig. 3. Trade Credit Scores and Bank Competition.

These results show that in tracts with relatively few branches, CRA eligibility has a strong positive effect on the average PAYDEX score 1 year after eligibility.

Turning to tracts with high bank competition, columns (5) and (6) of Table 6, Panel A, provide results for tracts with an HHI below 1800, while columns (7) and (8) present results for tracts with an above the median number of bank branches. Overall, these results indicate that the effect of CRA eligibility on small firms' PAYDEX scores is insignificant or statistically weaker than that found in low competition tracts.

Similarly, columns (1)–(4) of Table 6, Panel B, confirm that in tracts with low bank competition, CRA eligibility increases the proportion of firms with a PAYDEX score exceeding 70. However, columns (5)–(8) show that in high bank competition tracts, CRA eligibility tends not to have a statistically significant effect on raising the proportion of firms with a PAYDEX score over 70.

Taken together, the evidence supports the conclusion that small firms in high bank competition tracts are likely to receive bank loans even without added CRA inducements, so CRA eligibility does not significantly improve their trade credit performance.²⁴ In contrast, small firms located in tracts with low bank competition are

24. A univariate *t*-test of means shows that the average PAYDEX score prior to the change in CRA eligibility is higher among high bank competition tracts. Specifically, the average PAYDEX score in high bank competition tracts is 0.23 greater than that in low bank competition tracts, which is equivalent to five standard deviations, a statistically significant difference. Similarly, the proportion of firms with a PAYDEX

more likely to be unbanked and financially constrained. Thus, after their tracts become CRA eligible, these small firms are more likely to use their new bank loans to reduce their overdue trade credit.

3.4 Comparing Small Firms with Heterogeneous Trade Credit Dependence

Petersen and Rajan (1997) find that small firms in manufacturing and wholesale trade sectors display a higher dependence on trade credit compared to small firms in services and retail trade sectors. This section analyzes whether CRA-induced bank lending impacts trade credit differently across these two groups of firms. It examines whether small firms in sectors that are *ex-ante* more dependent on trade credit are more likely to use bank loans to improve their trade credit payment performance. Essentially, it uses heterogeneity in trade credit dependence to run a treatment versus placebo robustness test of our earlier findings.

We begin by calculating separately PAYDEX scores of firms in trade credit-dependent sectors and PAYDEX scores of firms not in these sectors, for both low and high bank competition census tracts. Figure 4 shows PAYDEX-related RD plots for firms in trade credit-dependent sectors by their tracts' MFI ratios. For low competition tracts, the left top (*bottom*) RD plot indicates a significant discontinuity in the average PAYDEX score (*the proportion of firms with a PAYDEX score exceeding 70*) at the 80% MFI ratio. However, consistent with earlier results for high competition tracts, the rightmost RD plots in Figure 4 reveal that the discontinuities in the average PAYDEX score and the proportion of firms with score higher than 70 vanish.

Figure 5 presents corresponding RD plots for firms in trade credit-independent sectors. Importantly, it reveals no statistically significant discontinuities in the average PAYDEX score or in the proportion of firms with scores exceeding 70, irrespective of the level of bank competition in the firms' census tracts. Taken together, the plot-based evidence in Figures 4 and 5 suggests a link between trade credit dependence and the documented effect that CRA-induced bank lending improves small firms' trade credit scores.

Table 7 presents more formal results on the effects that CRA-induced bank lending has on the trade credit performance of small firms in trade credit-dependent sectors. It provides strong evidence that only in low competition census tracts do firms in trade credit-dependent sectors use CRA-generated loans to pay their trade credit more promptly. Specifically, columns (1)–(2) of Table 7, Panel A, show that in low bank competition tracts defined by an HHI above 1800, there is a statistically and economically significant rise in PAYDEX scores among trade credit-dependent firms. The estimated PAYDEX improvement is up to 2.72 points, implying a 4.1-day decline in the lateness of their payments relative to invoice terms. When low competition tracts are defined as having below-median numbers of branches, columns (3)–(4) of

score exceeding 70 in high bank competition tracts is 0.65 percentage points greater than that in low bank competition tracts, which is equivalent to almost six standard deviations, a statistically significant difference.

TABLE 7
THE CRA'S EFFECT ON SMALL FIRMS IN TRADE CREDIT-DEPENDENT SECTORS

Variables	Panel A: PAYDEX score							
	Low bank competition tracts				High bank competition tracts			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Conventional	2.186** (0.889)	2.543*** (0.842)	3.080*** (1.021)	3.326*** (0.880)	1.235 (1.810)	1.466 (1.559)	0.637 (1.217)	0.485 (1.119)
Bias-corrected	2.286** (0.889)	2.718*** (0.842)	3.374*** (1.021)	3.576*** (0.880)	0.880 (1.810)	1.584 (1.559)	0.190 (1.217)	0.391 (1.119)
Robust	2.286** (1.025)	2.718*** (0.975)	3.374*** (1.203)	3.576*** (1.025)	0.880 (2.117)	1.584 (1.831)	0.190 (1.409)	0.391 (1.284)
Observations	49,404	10,280	40,187	8,171	20,909	3,538	33,002	6051
Robust 95% CI	[0.277; 4.296]	[0.807; 4.628]	[1.017; 5.732]	[1.567; 5.585]	[-3.27; 5.03]	[-2.004; 5.172]	[-2.572; 2.952]	[-2.125; 2.907]
Kernel Type	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
BW Type	mserd	mserd	mserd	mserd	mserd	mserd	mserd	mserd
Sample	Total	Matched	Total	Matched	Total	Matched	Total	Matched
Order Loc.Poly.(p)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Order Bias(q)	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
BW Loc.Poly.(h)	12.017	9.823	11.906	12.111	11.686	10.168	16.136	11.461
BW Bias (b)	21.796	17.787	20.035	21.401	19.615	16.937	29.440	21.115

(Continued)

TABLE 7
(CONTINUED)

Variables	Panel B: Proportion of firms with PAYDEX > 70							
	Low bank competition tracts				High bank competition tracts			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Conventional	0.0554** (0.0235)	0.0642*** (0.0211)	0.0602** (0.0260)	0.0751*** (0.0241)	0.00572 (0.0466)	0.00563 (0.0389)	0.0104 (0.0314)	0.0158 (0.0293)
Bias-corrected	0.0586** (0.0235)	0.0689*** (0.0211)	0.0633** (0.0260)	0.0755*** (0.0241)	0.00820 (0.0466)	0.0123 (0.0389)	0.00647 (0.0314)	0.0242 (0.0293)
Robust	0.0586** (0.0269)	0.0689*** (0.0242)	0.0633** (0.0305)	0.0755*** (0.0282)	0.00820 (0.0543)	0.0123 (0.0446)	0.00647 (0.0362)	0.0242 (0.0333)
Observations	49,404	10,280	40,187	8,171	20,909	3,538	33,002	6051
Robust 95% CI	[0.006; 0.111]	[0.022; 0.116]	[0.004; 0.123]	[0.02; 0.131]	[-0.098; 0.115]	[-0.075; 0.1]	[-0.064; 0.077]	[-0.041; 0.089]
Kernel Type	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
BW Type	mserd	mserd	mserd	mserd	mserd	mserd	mserd	mserd
Sample	Total	Matched	Total	Matched	Total	Matched	Total	Matched
Order Loc. Poly.(p)	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Order Bias(q)	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000
BW Loc. Poly.(h)	11,893	10,474	12,395	10,453	11,423	10,968	15,798	10,747
BW Bias (b)	21,846	19,800	21,097	17,947	18,912	19,315	28,843	20,387

NOTE: The outcome variables are the 1-year forward tract-level average PAYDEX score of small firms in trade credit-dependent sectors (Panel A) and the proportion of small firms in trade credit-dependent sectors with a PAYDEX score exceeding 70 (Panel B). Covariates include the tract's population, the population's proportion of minorities, the MFI, and the numbers of small and large bank branches. Bank competition is measured by HHI in (1) and (2) and by the number of bank branches in (3)–(4) and (7)–(8). The conventional estimator of RD is based on the first-order (linear) polynomial estimators in (4) and (5). Bias-corrected and robust is the first-order polynomial fuzzy RD estimator in (7). Reported bandwidths are percentage points of the MFI ratio. Standard errors in parentheses. Significance levels are ***1%, **5%, and *10%.

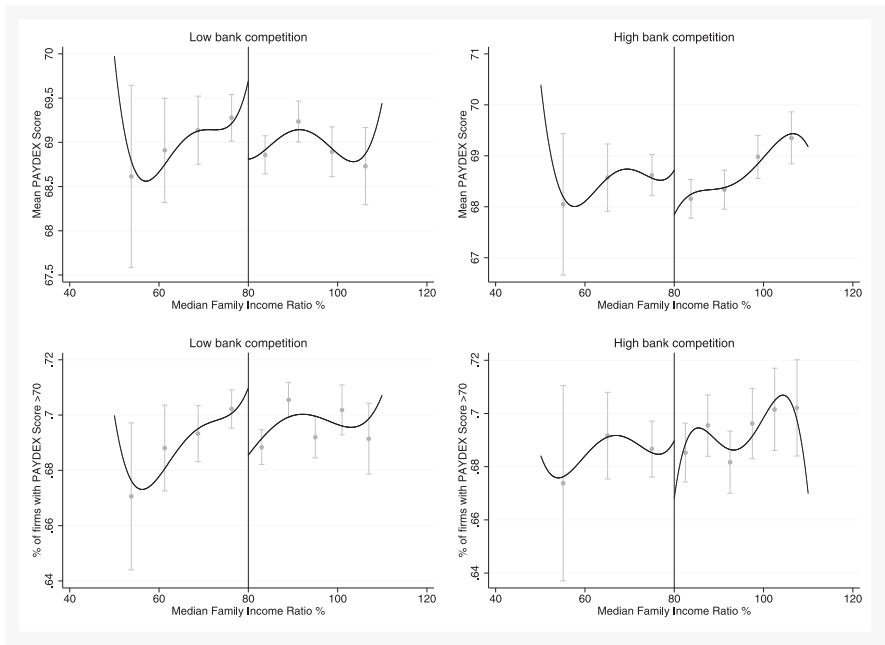


Fig. 4. Trade Credit Scores of Firms in Trade Credit-Dependent Sectors.

Table 7, Panel A, indicate an even greater improvement in PAYDEX, up to a 5.4-day reduction in time that payments are overdue. In contrast, columns (5)–(8) of Panel A show that in high competition tracts, there is no evidence of statistically significant declines in trade credit-dependent firms’ overdue payments.

Table 7, Panel B, provides qualitatively similar results to those in Panel A. Columns (1)–(4) show when trade credit-dependent firms operate in low bank competition tracts, CRA-induced loans raise the proportion firms with a PAYDEX exceeding 70. The estimated effect is statistically and economically significant and ranges from 5.9% to 7.6%. However, consistent with our earlier empirical evidence, columns (5)–(8) of Panel B indicate no statistically significant effect on trade credit when these firms operate in high bank competition tracts.

Table 8 repeats the estimation exercises presented in Table 7 but uses data on the subsample of small firms operating in trade credit-independent sectors rather than in trade credit-dependent sectors. Interestingly, it shows only weak or statistically insignificant evidence that CRA-induced bank lending leads to greater promptness in small firms’ payments to their trade creditors. Taken together, the results in Tables 7 and 8 show that CRA-generated bank loans significantly reduce the lateness of small firms’ payments to their trade creditors mainly when firms belong to industries

TABLE 8
THE CRA'S EFFECT ON SMALL FIRMS IN TRADE CREDIT-INDEPENDENT SECTORS

Variables	Panel A: PAYDEX score							
	Low bank competition tracts				High bank competition tracts			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Conventional	1.229* (0.669)	1.414** (0.641)	1.470* (0.827)	1.812*** (0.686)	1.873 (1.518)	2.234* (1.233)	1.338 (0.885)	1.800** (0.797)
Bias-corrected	1.101* (0.669)	1.282** (0.641)	1.126 (0.827)	1.656** (0.686)	2.119 (1.518)	2.303* (1.233)	1.311 (0.885)	1.827** (0.797)
Robust	1.101 (0.773)	1.282* (0.740)	1.126 (0.974)	1.656** (0.795)	2.119 (1.784)	2.303 (1.431)	1.311 (1.015)	1.827** (0.917)
Observations	49,521	10,298	40,286	8,187	20,952	3,543	33,070	6060
Robust 95% CI	[-0.414; 2.616]	[-0.169; 2.732]	[-0.784; 3.035]	[0.097; 3.215]	[-1.377; 5.616]	[-0.503; 5.108]	[-0.679; 3.3]	[0.03; 3.623]
Kernel Type	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
BW Type	mserd	mserd	mserd	mserd	mserd	mserd	mserd	mserd
Sample	Total	Matched	Total	Matched	Total	Matched	Total	Matched
Order Loc.Poly.(p)	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Order Bias(q)	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000
BW Loc.Poly.(h)	12.941	9.665	12.116	12.626	11.417	11.218	17.329	12.316
BW Bias (b)	23.600	17.553	20.522	22.890	18.942	19.331	33.434	22.335

(Continued)

TABLE 8
(CONTINUED)

Panel B: Proportion of firms with PAYDEX >70								
Variables	Low bank competition tracts			High bank competition tracts				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Conventional	0.0176 (0.0180)	0.0224 (0.0167)	0.0172 (0.0221)	0.0235 (0.0196)	0.0392 (0.0369)	0.0441 (0.0316)	0.0396* (0.0234)	0.0444** (0.0207)
Bias-corrected	0.0126 (0.0180)	0.0188 (0.0167)	0.00669 (0.0221)	0.0171 (0.0196)	0.0384 (0.0369)	0.0417 (0.0316)	0.0390* (0.0234)	0.0434** (0.0207)
Robust	0.0126 (0.0208)	0.0188 (0.0192)	0.00669 (0.0260)	0.0171 (0.0230)	0.0384 (0.0433)	0.0417 (0.0367)	0.0390 (0.0271)	0.0434* (0.0238)
Observations	49,521	10,298	40,286	8,187	20,952	3,543	33,070	6,060
Robust 95% CI	[−0.028; 0.053]	[−0.019; 0.056]	[−0.044; 0.058]	[−0.028; 0.062]	[−0.047; 0.123]	[−0.03; 0.114]	[−0.014; 0.092]	[−0.003; 0.09]
Kernel Type	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
BW Type	mserd	mserd	mserd	mserd	mserd	mserd	mserd	mserd
Sample	Total	Matched	Total	Matched	Total	Matched	Total	Matched
Order Loc.Poly.(p)	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
Order Bias(q)	2,000	2,000	2,000	2,000	2,000	2,000	2,000	2,000
BW Loc.Poly.(h)	12.677	10.228	12.098	10.765	12.453	11.162	16.681	12.682
BW Bias (b)	23,365	18,983	20,601	18,573	21,042	18,872	31,045	22,949

NOTE: The outcome variables are the 1-year forward tract-level average PAYDEX score of small firms in trade credit-independent sectors (Panel A) and the proportion of small firms in trade credit-independent sectors with a PAYDEX score exceeding 70 (Panel B). Covariates include the tract's population, the population's proportion of minorities, the MFI, and the numbers of small and large bank branches. Bank competition is measured by HHI (above versus below 1800) in (1)–(2) and by the number of bank branches (below versus above the median) in (3)–(4) and (7)–(8). The conventional estimator of RD is based on the first-order (linear) polynomial estimators in (4) and (5). Bias-corrected and robust is the first-order polynomial fuzzy RD estimator in (7). Reported bandwidths are percentage points of the MFI ratio. Standard errors in parentheses. Significance levels are ***1%, **5%, and *10%.

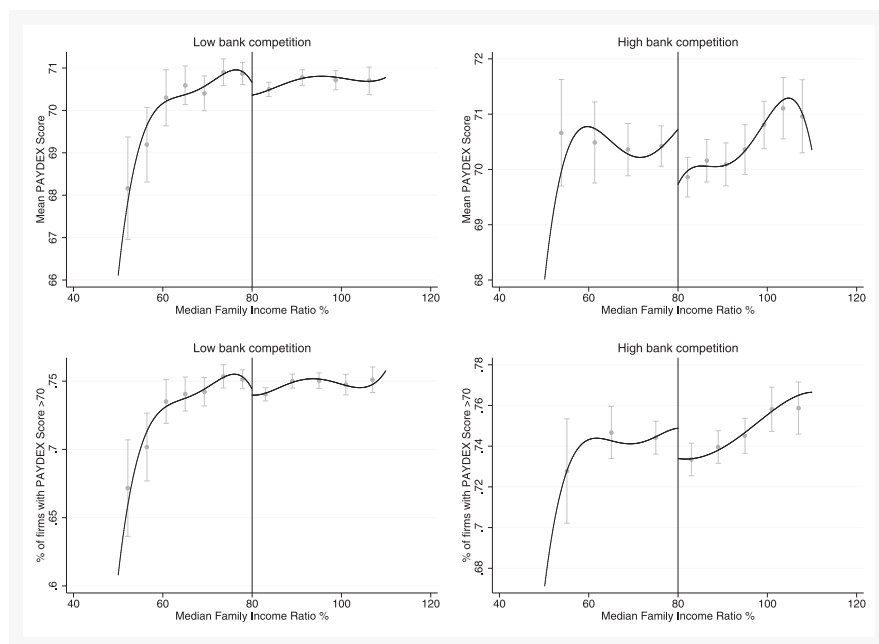


Fig. 5. Trade Credit Scores of Firms in Trade Credit-Independent Sectors.

that traditionally depend on trade credit and their local markets were served by few banks.

3.5 Falsification Analysis

Our RD design assigns CRA eligibility based on whether a census tract's MFI ratio is below a given 80% threshold. Assuming that a tract's MFI ratio is not manipulated by agents such as the tract's small firms or banks, then variation in CRA eligibility when the MFI ratio is near the threshold is random as in a randomized experiment (Lee and Lemieux 2010). Given the objective determination of this MFI-based assignment variable, a continuity assumption of the regression function near the cutoff threshold ensures the validity of the RD design. Although the continuity assumption cannot be directly tested, there are empirical tests that provide indirect evidence of its validity. Following Cattaneo, Idrobo, and Titiunik (2019), we consider three validation exercises based on (i) the insignificant eligibility effect on covariates (placebo outcomes), (ii) the eligibility effect observed at artificial cutoff values (placebo cutoffs), and (iii) the exclusion of observations too close to the cutoff (donut-hole analysis).

The first test notes that CRA eligibility should not be linked to any predetermined characteristic of a census tract. Since the eligibility effect on covariates is zero by

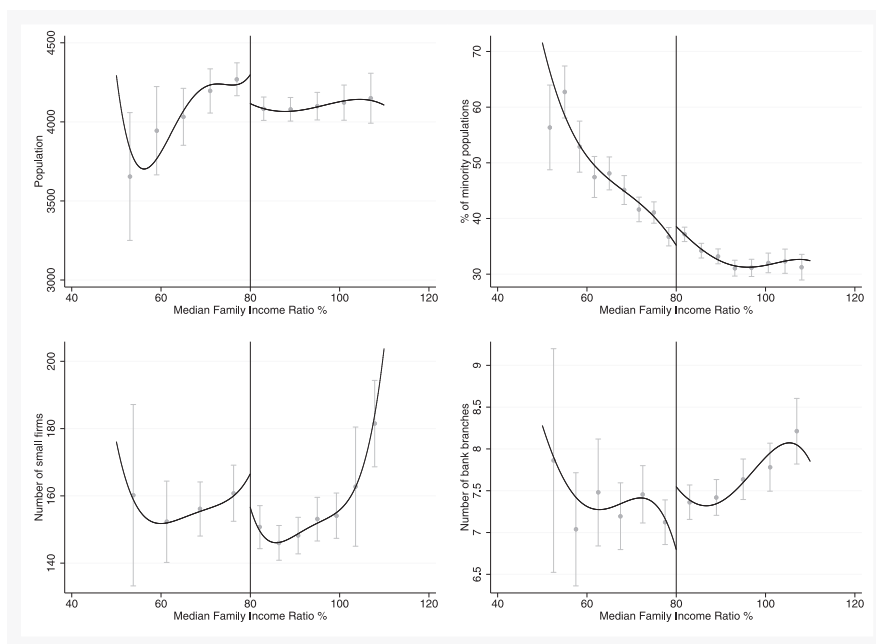


Fig. 6. Covariate (Dis)continuity at the 80% Tract MFI Ratio.

construction, consistent evidence of covariates' discontinuity would raise questions about the validity of the RD continuity assumption. For example, if tracts' population or the number of small firms were significantly different just above and below the eligibility cutoff, it could indicate a discontinuity in bank loan demand that could be an alternative explanation of the findings. Similarly, if tracts had significantly different numbers of bank branches at the eligibility cutoff, it would indicate a discontinuity in access to bank loans.

Therefore, we test for potential discontinuity jumps at the 80% tract MFI cutoff for our model covariates, namely, a tract's population, the population's proportion of minorities, the number of small firms, and the numbers of small and large bank branches. The RD plots in Figure 6 and the *p*-values of the CRA eligibility effect on the covariates in Table 9, columns (1)–(5), provide no evidence for statistically significant discontinuity jumps near the cutoff with respect to a tract's proportion of minorities, number of small firms, and the numbers of small and large bank branches. We find only weak evidence of a possible discontinuity for tract population: the robust estimates at the cutoff are different at the 10% significance level but not at 5% significance level.

Our second falsification exercise tests whether there are significant effects at placebo cutoff values. Evidence of discontinuity at MFI ratios away from the actual CRA cutoff would imply that CRA eligibility is not the only "treatment" effect,

TABLE 9
CONTINUITY ANALYSIS FOR COVARIATES

Variables	(1) Population	(2) Proportion of minorities	(3) No. of small firms	(4) No. of small banks	(5) No. of large banks
Conventional	$p = 0.0484$	$p = 0.703$	$p = 0.174$	$p = 0.384$	$p = 0.457$
Bias-corrected	$p = 0.0598$	$p = 0.784$	$p = 0.179$	$p = 0.382$	$p = 0.394$
Robust	$p = 0.0708$	$p = 0.791$	$p = 0.204$	$p = 0.408$	$p = 0.412$
Observations	73,366	73,366	73,366	73,366	73,366
Kernel Type	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
BW Type	cerdd	cerdd	cerdd	cerdd	cerdd
Order Loc.Poly.(p)	1.000	1.000	1.000	1.000	1.000
Order Bias(q)	2.000	2.000	2.000	2.000	2.000
BW Loc.Poly.(h)	6.384	5.882	6.964	6.428	4.606
BW Bias (b)	20.408	19.886	16.948	17.651	14.362

NOTE: The Conventional estimator p -value is based on the first-order (linear) polynomial estimators in (4) and (5), and bias-corrected and robust p -values are based on the first-order polynomial RD estimator in (6) with outcome variables the covariates of the tract's population, the population's proportion of minorities, the number of small firms, and the numbers of small and large bank branches. Reported bandwidths are percentage points of the MFI ratio.

casting doubt on our RD design. Therefore, we repeat the RD estimation but replace the MFI ratio's true cutoff value of 80% with placebo cutoff ratios of 70% and 90% where CRA eligibility does not change. Columns (1) and (4) of Table 10 show that at these placebo cutoff values, the estimated effects on the probabilities of a substantial increase in bank loans to small firms are not generally statistically significant. Similarly, at the placebo cutoffs, there are no statistically significant changes in small firms' PAYDEX scores (columns (2) and (5)) or in the proportion of small firms with a PAYDEX score exceeding 70 (columns (3) and (6)). Therefore, we find no evidence of a discontinuity in small firms' trade credit scores beyond the one induced by CRA eligibility at the true 80% MFI ratio.

Our third and final falsification exercise examines whether our results are sensitive to a few census tracts located very close to the 80% MFI ratio determining CRA eligibility. It tests whether the results are robust to the unavoidable extrapolation involved in local polynomial estimation, as the tract observations nearest the cutoff are more influential since they receive higher kernel weights. If the exclusion of those tracts changes the significance of the results, then the external validity of the findings is in doubt. Excluding such tracts and then repeating the estimation using the remaining sample is sometimes referred to as a "donut hole" approach. Accordingly, we remove tracts within a radius of 2% from the 80% MFI ratio cutoff for CRA eligibility, which reduces the full sample of census tracts by 3.4%. The results in Table 11 show that the estimated CRA-eligibility effects on the probability of a substantial increase in bank loans to small firms, on firms' average PAYDEX score, and on the proportion of firms with a PAYDEX score exceeding 70 remain statistically significant and economically equivalent to the main results.

TABLE 10
PLACEBO CUTOFF ANALYSIS

TABLE 10 PLACEBO CUTOFF ANALYSIS						
Variables	Cutoff 70%			Cutoff 90%		
	(1) Bank loans	(2) PAYDEX score	(3) % PAYDEX > 70	(4) Bank loans	(5) PAYDEX score	(6) % PAYDEX > 70
Conventional	-0.0837 ^{**} (0.0416)	7.124 (5.294)	0.126 (0.113)	0.000781 (0.0145)	-552.0 (32,539)	-216.0 (131,072)
Bias-corrected	-0.0687 [*] (0.0416)	7.004 (5.294)	0.122 (0.113)	0.00274 (0.0145)	-24,588 (32,539)	-87,420 (131,072)
Robust	-0.0687 (0.0473)	7.004 (6.024)	0.122 (0.127)	0.00274 (0.0174)	-24,588 (34,250)	-87,420 (143,173)
Observations	73,366	73,366	73,366	73,366	73,366	73,366
Kernel Type	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
BW Type	mserd	mserd	mserd	mserd	mserd	mserd
Sample	Total	Total	Total	Total	Total	Total
Order Loc.Poly.(p)	1,000	1,000	1,000	1,000	1,000	1,000
Order Bias(q)	2,000	2,000	2,000	2,000	2,000	2,000
BW Loc.Poly.(h)	7,833	9,231	9,143	5,370	6,170	5,825
BW Bias (b)	14,549	17,487	18,260	8,580	18,384	13,732

NOTE: The outcome variables are the probability of an increase of at least 20% in either the number or the total amount of bank loans made to small firms, the 1-year forward tract-level average PAYDEX score of small firms, and the proportion of small firms with a PAYDEX score exceeding 70. Covariates include the tract's population, the population's proportion of minorities, the MFI, and the numbers of small and large bank branches. The conventional estimator of RD is based on the first-order (linear) polynomial estimators in (4) and (5). Bias-corrected and robust are the first-order polynomial RD estimators in (6) (columns (1) and (4)) and in (7) (columns (2)–(3) and (5)–(6)). Alternative cutoff points of the MFI ratio are 70% and 90%. Standard errors in parentheses. Reported bandwidths are percentage points of the MFI ratio. Significance levels are ***1%, **5%, and * 10%.

TABLE 11
DONUT HOLE ANALYSIS USING RATIOS OF MFI < 78% AND MFI > 82%

Variables	(1) Bank loans	(2) Bank loans	(3) PAYDEX score	(4) PAYDEX score	(5) % PAYDEX > 70	(6) % PAYDEX > 70
Conventional	0.213*** (0.0360)	0.309*** (0.0363)	2.584*** (0.805)	2.239** (0.885)	0.0630*** (0.0201)	0.0583*** (0.0222)
Bias-corrected	0.216*** (0.0360)	0.312*** (0.0363)	2.589*** (0.805)	2.274** (0.885)	0.0608*** (0.0201)	0.0604*** (0.0222)
Robust	0.216*** (0.0442)	0.312*** (0.0443)	2.589*** (0.965)	2.274** (1.072)	0.0608** (0.0239)	0.0604** (0.0268)
Observations	70,769	12,554	70,769	12,554	70,769	12,554
Kernel Type	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
BW Type	mserd	mserd	mserd	mserd	mserd	mserd
Sample	Total	Matched	Total	Matched	Total	Matched
Order Loc.Poly.(p)	1,000	1,000	1,000	1,000	1,000	1,000
Order Bias(q)	2,000	2,000	2,000	2,000	2,000	2,000
BW Loc.Poly.(h)	10,317	10,653	15,226	10,847	16,089	11,162
BW Bias (b)	22,219	23,411	36,664	23,962	40,693	24,798

NOTE: The outcome variables are the probability of an increase of at least 20% in either the number or the total amount of bank loans made to small firms, the 1-year forward tract-level average PAYDEX score of small firms, and the proportion of small firms with a PAYDEX score exceeding 70. Covariates include the tract's population, the population's proportion of minorities, the MFI, and the numbers of small and large bank branches. The conventional RD estimator is based on the first-order (linear) polynomial estimators in (4) and (5). Bias-corrected and robust are the first-order polynomial RD estimators in (6) (columns (1) and (2)) and in (7) (columns (3)–(6)). Tracts with an MFI ratio within a 2% radius around the 80% cutoff are excluded from the sample. Reported bandwidths are percentage points of the MFI ratio. Standard errors in parentheses. Significance levels are ***, **, *, and *10%.

4. CONCLUSION

This study presented empirical evidence that when a census tract becomes newly eligible for CRA lending, there is a higher probability of a substantial increase in bank lending to small firms in that tract compared to tracts that remain ineligible. However, only in newly-eligible tracts with low bank competition do small firms significantly improve the promptness of their payments to trade creditors. The improvement is especially pronounced when a small firm operates in an industry that traditionally relied on trade credit. Our study confirms past research showing that the CRA tends to expand bank lending to small firms in LMI areas. But it also finds that the CRA can economically benefit small firms by providing bank credit that reduces the lateness of their trade credit payments and potentially expensive delinquency charges.

These findings are consistent with the notion that small firms operating in local markets with many banks are less likely to suffer from trade payment difficulties because they are receiving adequate bank credit. In contrast, in the absence of government regulation, small firms in markets with few banks are likely to be financially constrained, resulting in late payments to their trade creditors. Government intervention may be needed to stimulate bank lending to these unbanked or underbanked small firms. One way to accomplish this goal is through CRA-type regulations that incentivize banks to expand lending to LMI areas. While LMI areas tend to have higher bank concentration than UMI areas, the regulation might be most effective if it focuses rewards for lending in those areas with less bank competition. As an alternative, or in addition, to CRA regulation, government interventions might take the form of policies that promote greater bank competition in LMI areas that are currently served by few banks. For example, encouraging the entry of bank branches in underserved areas might promote greater competition in lending to small firms.

Although our study sheds new light on the relationship between bank lending and trade credit, as well as the relationship between CRA regulation and competition, it raises a number of other interesting issues for future research. Further investigation may be warranted to determine the CRA's effect on the suppliers of trade credit. The CRA could benefit a small firm's vendors and suppliers by shortening the trade credit period, though its ultimate effect needs to account for potential losses from early-payment discounts or from the elimination of late-payment fees. Furthermore, our paper has focused on bank competition with trade credit being a firm's only source of nonbank financing. Yet, the recent growth in online marketplace lending offers an additional source of funding that could strengthen competition and reduce the need for the CRA.

LITERATURE CITED

- Agarwal, Sumit, and Robert Hauswald. (2010) "Distance and Private Information in Lending." *Review of Financial Studies*, 23, 2757–88.

- Amberg, Niklas, Tor Jacobson, Erik Von Schedvin, and Robert Townsend. (2021) "Curbing Shocks to Corporate Liquidity: The Role of Trade Credit." *Journal of Political Economy*, 129, 182–242.
- Avery, Robert B., and Kenneth P. Brevoort. (2015) "The Subprime Crisis: Is Government Housing Policy to Blame?" *Review of Economics and Statistics*, 97, 352–63.
- Avery, Robert B., Paul S. Calem, and Glenn B. Canner. (2003) "The Effects of the Community Reinvestment Act on Local Communities." Board of Governors of the Federal Reserve System Working Paper.
- Avery, Robert B., Raphael W. Bostic, and Glenn B. Canner. (2005) "Assessing the Necessity and Efficiency of the Community Reinvestment Act." *Housing Policy Debate*, 16, 143–72.
- Barnatchez, Keith, Leland D. Crane, and Ryan A. Decker. (2017) "An Assessment of the National Establishment Time Series (NETS) Database." Finance and Economics Discussion Series 2017-110, Federal Reserve Board.
- Barrot, Jean-Noël. (2016) "Trade Credit and Industry Dynamics: Evidence from Trucking Firms." *Journal of Finance*, 71, 1975–2016.
- Begley, Taylor A., and Amiyatosh Purnanandam. (2021) "Color and Credit: Race, Regulation, and the Quality of Financial Services." *Journal of Financial Economics*, 141, 48–65.
- Berger, Allen N., and Gregory F. Udell. (1998) "The Economics of Small Business Finance: The Roles of Private Equity and Debt Markets in the Financial Growth Cycle." *Journal of Banking & Finance*, 22, 613–73.
- Berger, Allen N., Anthony Saunders, Joseph M. Scalise, and Gregory F. Udell (1998) "The Effects of Bank Mergers and Acquisitions on Small Business Lending." *Journal of Financial Economics*, 50, 187–229.
- Berger, Allen N., Nathan H. Miller, Mitchell A. Petersen, Raghuram G. Rajan, and Jeremy C. Stein. (2005) "Does Function Follow Organizational Form? Evidence from the Lending Practices of Large and Small Banks." *Journal of Financial Economics*, 76, 237–69.
- Bhutta, Neil. (2011) "The Community Reinvestment Act and Mortgage Lending to Lower Income Borrowers and Neighborhoods." *Journal of Law and Economics*, 54, 953–83.
- Biais, Bruno, and Christian Gollier. (1997) "Trade Credit and Credit Rationing." *Review of Financial Studies*, 10, 903–37.
- Black, Sandra E., and Philip E. Strahan. (2002) "Entrepreneurship and Bank Credit Availability." *Journal of Finance*, 57, 2807–33.
- Bostic, Raphael W., and Hyojung Lee. (2017) "Small Business Lending under the Community Reinvestment Act." *Cityscape*, 19, 63–84.
- Bostic, Raphael W., and Breck L. Robinson. (2003) "Do CRA Agreements Influence Lending Patterns?" *Real Estate Economics*, 31, 23–51.
- Brevoort, Kenneth, John Holmes, and John Wolken. (2010) "Distance Still Matters: The Information Revolution in Small Business Lending and the Persistent Role of Location, 1993–2003." Finance and Economics Discussion Series Working Paper 2010-08, Federal Reserve Board.
- Breza, Emily, and Andres Liberman. (2017) "Financial Contracting and Organizational Form: Evidence from the Regulation of Trade Credit." *Journal of Finance*, 72, 291–324.
- Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik. (2014) "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs." *Econometrica*, 82, 2295–326.
- Cattaneo, Matias D., Nicolás Idrobo, and Rocio Titiunik. (2019) *A Practical Introduction to Regression Discontinuity Designs: Foundations*. Cambridge: Cambridge University Press.

- Cetorelli, Nicola, and Philip E. Strahan. (2006) "Finance as a Barrier to Entry: Bank Competition and Industry Structure in Local US Markets." *Journal of Finance*, 61, 437–61.
- Cunat, Vicente. (2006) "Trade Credit: Suppliers as Debt Collectors and Insurance Providers." *Review of Financial Studies*, 20, 491–527.
- Cunat, Vicente, and Emilia Garcia-Appendini. (2012) "Trade Credit and Its Role in Entrepreneurial Finance." In *Oxford Handbook of Entrepreneurial Finance*, edited by Douglas Cumming, pp. 526–57. New York: Oxford University Press.
- Dahl, Drew, Douglas D. Evanoff, and Michael F. Spivey. (2010) "The Community Reinvestment Act and Targeted Mortgage Lending." *Journal of Money, Credit and Banking*, 42, 1351–72.
- Ding, Lei, and Leonard Nakamura. (2020) "Don't Know What You Got till It's Gone - The Community Reinvestment Act in a Changing Financial Landscape." Working Papers WP 20-08, Federal Reserve Bank of Philadelphia.
- Ding, Lei, Hyojung Lee, and Raphael Bostic. (2018) "Effects of the Community Reinvestment Act (CRA) on Small Business Lending." Working Paper WP 18-27, Federal Reserve Bank of Philadelphia.
- Fisman, Raymond, and Inessa Love. (2003) "Trade Credit, Financial Intermediary Development, and Industry Growth." *Journal of Finance*, 58, 353–74.
- Giannetti, Mariassunta, Mike Burkart, and Tore Ellingsen. (2011) "What You Sell Is What You Lend? Explaining Trade Credit Contracts." *Review of Financial Studies*, 24, 1261–98.
- Giannetti, Mariassunta, Nicolas Serrano-Velarde, and Emanuele Tarantino. (2021) "Cheap Trade Credit and Competition in Downstream Markets." *Journal of Political Economy*, 129, 1744–96.
- Greenstone, Michael, Alexandre Mas, and Hoai-Luu Nguyen. (2020) "Do Credit Market Shocks Affect the Real Economy? Quasi-Experimental Evidence from the Great Recession and "Normal" Economic Times." *American Economic Journal: Economic Policy*, 12, 200–25.
- Heckman, James J., Robert J. LaLonde, and Jeffrey A. Smith. (1999) "The Economics and Econometrics of Active Labor Market Programs." In *Handbook of Labor Economics*, edited by Orley C. Ashenfelter and David Card, pp. 1865–2097. Amsterdam: Elsevier.
- Jacobson, Tor, and Erik Von Schedvin. (2015) "Trade Credit and the Propagation of Corporate Failure: An Empirical Analysis." *Econometrica*, 83, 1315–71.
- Kallberg, Jarl G., and Gregory F. Udell. (2003) "The Value of Private Sector Business Credit Information Sharing: The US Case." *Journal of Banking and Finance*, 27, 449–69.
- Kim, Mee Jung, Kyung Min Lee, and John S. Earle. (2021) "Does the Community Reinvestment Act Increase Lending to Small Businesses in Lower Income Neighborhoods?" *Economics Letters*, 209, 110–46.
- Klapper, Leora, Luc Laeven, and Raghuram Rajan. (2012) "Trade Credit Contracts." *Review of Financial Studies*, 25, 838–67.
- Lee, David S., and Thomas Lemieux. (2010) "Regression Discontinuity Designs in Economics." *Journal of Economic Literature*, 48, 281–355.
- Kramer Mills, Claire, Jessica Battisto, and Scott Lieberman. (2019) "Growing Pains: Examining Small Business Access to Affordable Credit in Low-Income Areas." *Consumer & Community Content*, 1, 22–7.
- Murfin, Justin, and Ken Njoroge. (2015) "The Implicit Costs of Trade Credit Borrowing by Large Firms." *Review of Financial Studies*, 28, 112–45.

- Ng, Chee K., Janet Kiholm Smith, and Richard L. Smith. (1999) "Evidence on the Determinants of Credit Terms Used in Interfirm Trade." *Journal of Finance*, 54, 1109–29.
- Nguyen, Hoai, -Luu Q. (2019) "Are Credit Markets Still Local? Evidence from Bank Branch Closings." *American Economic Journal: Applied Economics*, 11, 1–32.
- Petersen, Mitchell A., and Raghuram G. Rajan. (1994) "The Benefits of Lending Relationships: Evidence from Small Business Data." *Journal of Finance*, 49, 3–37.
- Petersen, Mitchell A., and Raghuram G. Rajan. (1995) "The Effect of Credit Market Competition on Lending Relationships." *Quarterly Journal of Economics*, 110, 407–43.
- Petersen, Mitchell A., and Raghuram G. Rajan. (1997) "Trade Credit: Theories and Evidence." *Review of Financial Studies*, 10, 661–91.
- Petersen, Mitchell A., and Raghuram G. Rajan. (2002) "Does Distance Still Matter? The Information Revolution in Small Business Lending." *Journal of Finance*, 57, 2533–70.
- Rice, Tara, and Philip E. Strahan. (2010) "Does Credit Competition Affect Small-Firm Finance?" *Journal of Finance*, 65, 861–89.
- Zinman, Jonathan. (2002) "The Efficacy and Efficiency of Credit Market Interventions: Evidence from the Community Reinvestment Act." Joint Center for Housing Studies of Harvard University Working Paper CRA02-2.